Nuclear Fission Plant

Sustainable Nuclear Energy

Assignment 3

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Summary

1 General introduction

In order to evaluate the thermo-mechanical constraints at critical points in nuclear fission systems, a detailed analysis is carried out that combined the effects of internal pressure, thermal gradients, and structural properties of the materials involved. This assessment is carried out according to the criteria set by ASME Section III, which provides the framework for verifying structural integrity under normal and accidental operating conditions. Special attention is devoted to fuel cladding regions, where combined mechanical and thermal stresses may approach the material limits, potentially compromising the safety margins. An evaluation of the gas plenum is also performed, as it plays a fundamental role in mitigating the internal pressure that could otherwise exceed the structural limits of the fuel pin.

2 Assumptions and data

To simplify the analysis, the following assumptions are adopted:

- The reactor operates under steady-state nominal conditions;
- Only static loads are taken into account, while dynamic effects are neglected;
- Bending stresses, peak stresses, and local stress concentrations are not considered;
- A 20% overpower margin is applied when calculating thermal stresses;
- The cladding material is assumed to be Zircaloy-4;
- The maximum internal pressure is evaluated under the conservative assumption of complete reactor depressurization ($P_o \approx 0$), and this value is used in the mechanical stress calculations for the cladding.

The principal operating parameters and geometrical data are taken from the *AP1000 Design Control Document*, while additional information (such as the inner cladding temperature) is derived from the results of Assignment 2. The yield strength and the ultimate tensile strength of the cladding material are assumed to be $\sigma_y = 241\,\text{MPa}$ and $\sigma_u = 413\,\text{MPa}$, respectively.

3 Mechanical integrity and cladding design

In accordance with Assignment 2, the scripts were developed in MATLAB, importing the necessary data from the previous assignment through a `.mat` file, used in a similar manner to a `.csv` file.

In this section, the steps for studying mechanical and thermal stresses are explained, and in the next section the results will be discussed.

3.1 Preliminary verification against buckling

It is important to evaluate the maximum pressure that the cladding can withstand before it undergoes structural deformation due to buckling. This critical pressure is computed using Equation 1:

$$P_{cr} = \frac{E}{2\,(1-\nu)} \left(\frac{t}{r_{avg}} \right)^3 \tag{1}$$

Where E is the Young's modulus (Equation 2), ν is Poisson's ratio (Equation 3), t is the cladding thickness, and r_{avg} is the average distance between the fuel pellet centerline and the cladding.

$$E = [9.9 \times 10^3 - 5.669(T - 273)] \times 9.81 \tag{2}$$

$$\nu = 0.3303 + 8.376 \times 10^{-5}(T - 273) \tag{3}$$

All properties of the material (E, ν) [1] are evaluated at the average temperature between the inner and outer surfaces at each axial point of the cladding, producing an array of critical pressure. Then it is token the minimum value to evaluate the worst-case scenario, which coincides with the peak of temperature profile, as it is possible to see in the Figure 1.

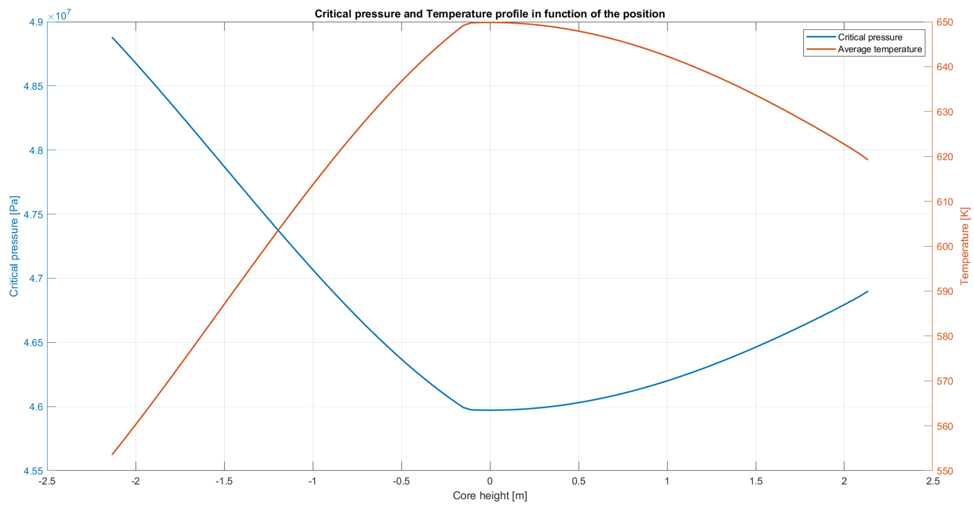

Figure 1: Critical pressure and Temperature profile in function of the position

[1] Ref: Kye-Ho, Nho. "Effect of high temperature steam oxidation on yielding of Zircaloy-4 PWR fuel cladding-expanding copper mantest." Nuclear Engineering and Technology 21.2 (1989): 111-122.

P_{cr} represents the critical pressure that the cladding can sustain before buckling occurs. Consequently, it is compared to the operating pressure acting on the cladding under conservative conditions.

3.2 Internal pressure

The maximum internal pressure that the Zircaloy-4 cladding can support before reaching its yield strength is evaluated. It is calculated using Mariotte's Equation 4, by replacing the hoop stress (σ_h) with the yield strength (σ_y).

$$P_{i,max} = \frac{\sigma_h \cdot t}{r_{avg}} \tag{4}$$

This value represents the limiting condition beyond which plastic deformation of the material would occur, potentially compromising the structural integrity of the fuel rod.

3.3 Verification of primary and secondary stresses with the Tresca criterion

The Tresca criterion is applied to assess whether the coating can withstand thermal and mechanical stresses under normal operating conditions, as well as in a scenario of power exceedance, where the thermal power reaches approximately $P_{\text{max}} \approx 1.2 P_{\text{th}}$. Usually the evaluation of the thermal stress is not made with over-power, but in this study the aim is to evidence the thermal effects, otherwise it could be difficult to enhance those. First, each component of the mechanical stresses (P_m) is evaluated for both the inner and outer surfaces of the cladding, using the following equations:

$$\text{Inner wall:} \quad \begin{cases} \sigma_h = \dfrac{P_i(r_o^2 + r_i^2) - 2P_o r_o^2}{r_o^2 - r_i^2} & \text{(5a)} \\ \sigma_r = -P_i & \text{(5b)} \\ \sigma_\ell = \dfrac{r_i^2(P_i - P_o)}{r_o^2 - r_i^2} & \text{(5c)} \end{cases}$$

$$\text{Outer wall:} \quad \begin{cases} \sigma_h = \dfrac{2P_o r_o^2 - P_i(r_o^2 + r_i^2)}{r_o^2 - r_i^2} & \text{(6a)} \\ \sigma_r = -P_o & \text{(6b)} \\ \sigma_\ell = \dfrac{r_i^2(P_i - P_o)}{r_o^2 - r_i^2} & \text{(6c)} \end{cases}$$

Where P is the pressure evaluated inside the road (P_i) and in the subchannel (P_o), while r_o and r_i are respectively the inner and outer radius of the cladding evaluated at cold condition, it means without thermal expansion of radius. For the evaluation of thermal stresses (Q), a 20% overpower condition is considered. The corresponding temperature, obtained from the script developed in Assignment 2, is used to compute the thermal stresses according to the following equations:

$$\text{Inner wall:}\quad \begin{cases} \sigma_{th,h} = \dfrac{E\alpha\left(T_{o,cl}-T_{i,cl}\right)}{2\left(1-\nu\right)}\dfrac{\left(1-\frac{2r_o^2}{r_o^2-r_i^2}\ln\left(\frac{r_o}{r_i}\right)\right)}{\ln\left(\frac{r_o}{r_i}\right)} & \text{(7a)} \\ \sigma_{th,r} = 0 & \text{(7b)} \\ \sigma_{th,\ell} = \dfrac{E\alpha\left(T_{o,cl}-T_{i,cl}\right)}{2\left(1-\nu\right)}\dfrac{\left(1-\frac{2r_o^2}{r_o^2-r_i^2}\ln\left(\frac{r_o}{r_i}\right)\right)}{\ln\left(\frac{r_o}{r_i}\right)} & \text{(7c)} \end{cases}$$

$$\text{Outer wall:}\quad \begin{cases} \sigma_{th,h} = \dfrac{E\alpha\left(T_{o,cl}-T_{i,cl}\right)}{2\left(1-\nu\right)}\dfrac{\left(1-\frac{2r_i^2}{r_o^2-r_i^2}\ln\left(\frac{r_o}{r_i}\right)\right)}{\ln\left(\frac{r_o}{r_i}\right)} & \text{(8a)} \\ \sigma_r = 0 & \text{(8b)} \\ \sigma_\ell = \dfrac{E\alpha\left(T_{o,cl}-T_{i,cl}\right)}{2\left(1-\nu\right)}\dfrac{\left(1-\frac{2r_i^2}{r_o^2-r_i^2}\ln\left(\frac{r_o}{r_i}\right)\right)}{\ln\left(\frac{r_o}{r_i}\right)} & \text{(8c)} \end{cases}$$

Where α[1] is the thermal expansion coefficient $\left[\frac{1}{K}\right]$, $T_{o,cl}$ and $T_{i,cl}$ are the temperature of the inner wall of the cladding and the outer wall of the cladding respectively $[K]$. In both cases, the radial stress component is assumed to be zero, because the cladding is unconstrained in the radial direction.

Using these sets of Equations (5)(6)(7)(8), for each case it is possible to evaluate the maximum stress acting on the cladding. Tresca's criterion computes the equivalent stress as:

$$\tau_{max} = max\left\{\frac{|\sigma_h-\sigma_r|}{2};\frac{|\sigma_r-\sigma_\ell|}{2};\frac{|\sigma_h-\sigma_\ell|}{2}\right\} \tag{9}$$

The value of τ_{max} is evaluated for the mechanical stress P_m and for the sum of thermal and mechanical stress P_m+Q. In according with ASME code section III, for the mechanical side, the primary stress τ_{max} is compared with the allowable stress (σ_a).

$$2\tau_{max} = \sigma_{max} \leq \sigma_a = S_m \tag{10}$$

Where S_m is evaluated as $min\left(\frac{2}{3}\sigma_y;\frac{1}{3}\sigma_u\right)$. While the τ_{max} come from the sum of primary and secondary stress must be lower than 3 times of S_m.

$$P_m+Q \leq \sigma_a = 3S_m \tag{11}$$

3.4 Gas plenum

The importance of the gas plenum is given by the ability to manage the gases produced by fission if designed properly. The gas plenum is useful for the storage of fission gases, such as H_2O and N_2, the optimal design is important to not exceed the physical limits due to the mechanical properties of the cladding, it is used to contain the pressure inside the fuel rod.

At the end of life (EOF), the fission gases are evaluated starting by computing the moles of gas impurities.

$$n_g = \frac{c_g \cdot M_{UO_2}}{u_g} \tag{12}$$

In which c_g is the concentration of gas impurities, M_{UO_2} represents the total mass of $\mathrm{UO_2}$ in the pin and u_g is the molecular weight of gas impurity. Subsequently, the number of moles of the fission gases ($n_{\mathrm{Xe+Kr}}$) is computed using:

- N_f: total number of fissions occurred in the pin;
- BU: burnup $\left[\frac{MW \cdot d}{kg}\right]$;
- E_f: energy released per fission $[MW \cdot d]$;
- M_U: mass of uranium $[kg]$;
- M_{UO_2}: mass of uranium dioxide $[kg]$;
- u_x: molecular weight of "x";
- R_r: fuel gas release ratio;
- N_A: Avogadro's number;
- Y: fission yield for Xe + Kr.

Firstly is it evaluated the mass of uranium as:

$$M_{\mathrm{U}} = M_{\mathrm{UO_2}} \cdot \frac{u_{\mathrm{U}}}{u_{\mathrm{UO_2}}} \tag{13}$$

This value is useful to compute the number of fission occurred in the pin:

$$N_f = \frac{BU \cdot M_{\mathrm{U}}}{E_f} \tag{14}$$

Finally it is possible compute $n_{\mathrm{Xe+Kr}}$ as:

$$n_{\mathrm{Xe+Kr}} = \frac{N_f \cdot Y \cdot R_r}{N_A} \tag{15}$$

After adding the number of moles evaluated in the Equation 15 and the moles of $\mathrm{H_2O}$ and $\mathrm{N_2}$, finally, it is possible compute the minimum value of volume of gas plenum through the law of the ideal gas as:

$$V_{min} = \frac{n \cdot R \cdot T}{p_{max}} \tag{16}$$

In this expression: n denotes the total number of moles; R is the constant of the ideal gases $\left[\frac{L \cdot Pa}{mol \cdot K}\right]$; T represents the temperature of the gases $[K]$; and p_{max} is the maximum pressure the cladding can support $[Pa]$.

4 Discussion of the results

ASME classifies loading condition as:

A) Service condition limits normal;

B) Service condition limits upset;

C) Service condition limits emergency;

D) Service condition limits faulted;

– Test condition limits;

In this treatment the evaluations have been made referring to design conditions and level service A, this means the pressure of the coolant is the nominal pressure (155 bar) and the pressure inside the cladding is evaluated as the maximum value the material can support (Equation 4) when the external pressure is equal to zero (in case of LOCA).

As shown in the Figure 1, the maximum value of the temperature corresponds to the minimum value of the critical pressure. This means that the maximum temperature corresponds to the worse condition; for this reason, the maximum value of the inner wall of the cladding was taken into account. The preliminary buckling verification is satisfied, as the minimum value of the critical pressure is 45.97 MPa, which is approximately three times greater than the operating internal pressure.

By applying Equation 4, the maximum pressure beyond which the cladding begins to lose its mechanical integrity is found to be 30.85 MPa.

Each stress value is plotted in the following two tables, divided by inner stress(Table 1) and outer stress (Table 2).

Inner stresses

	σ_h	σ_r	σ_l	σ_{Tresca}
Pm	104.80 MPa	-30.85 MPa	52.49 MPa	135.66 MPa
Q	28.75 MPa	0.00 MPa	28.75 MPa	28.75 MPa
Pm+Q	133.56 MPa	-30.85 MPa	81.24 MPa	164.41 MPa

Table 1: Inner stress

Outer stresses

	σ_h	σ_r	σ_l	σ_{Tresca}
Pm	89.46 MPa	-15.51 MPa	52.49 MPa	104.98 MPa
Q	-27.54 MPa	0.00 MPa	-27.54 MPa	27.54 MPa
Pm+Q	61.92 MPa	-15.51 MPa	24.94 MPa	77.43 MPa

Table 2: Outer stress

According to Equation 10, the mechanical stresses are compared to the allowable stress limit S_m which is equal to 137.7 MPa. The limit is respected for both value (inner and outer stress).

$$\begin{aligned}
&\text{[Inner stress]} \\
&\qquad \sigma_{\max} = 137.66 \text{ MPa} < S_m = 137.67 \text{ MPa} \\
&\text{[Outer stress]} \\
&\qquad \sigma_{\max} = 104.98 \text{ MPa} < S_m = 137.67 \text{ MPa}
\end{aligned}$$

Also the Equation 11 is verified:

$$\begin{aligned}
&\text{[Inner stress]} \\
&\qquad \sigma_{\max} = 164.41 \text{ MPa} < 3 \cdot S_m = 413.01 \text{ MPa} \\
&\text{[Outer stress]} \\
&\qquad \sigma_{\max} = 77.43 \text{ MPa} < 3 \cdot S_m = 413.01 \text{ MPa}
\end{aligned}$$

Every equation is verified, this means that the reactor operates within the physical limits of the cladding, despite the thermal evaluation being performed with an overpower.

Finally, the required gas plenum size was determined to be 20.54 cm, at this value it should be added the length of the spring, 15 cm. It is important to verify the ideal-gas assumption, it is possible using the compressibility factor Z, which was calculated as:

$$Z = \frac{n\,RT}{p\,V}$$

The closer Z is to one, the more valid the ideal-gas approximation. In this case, $Z = 0.983$, which confirms that treating water vapor and nitrogen as ideal gases is justified.